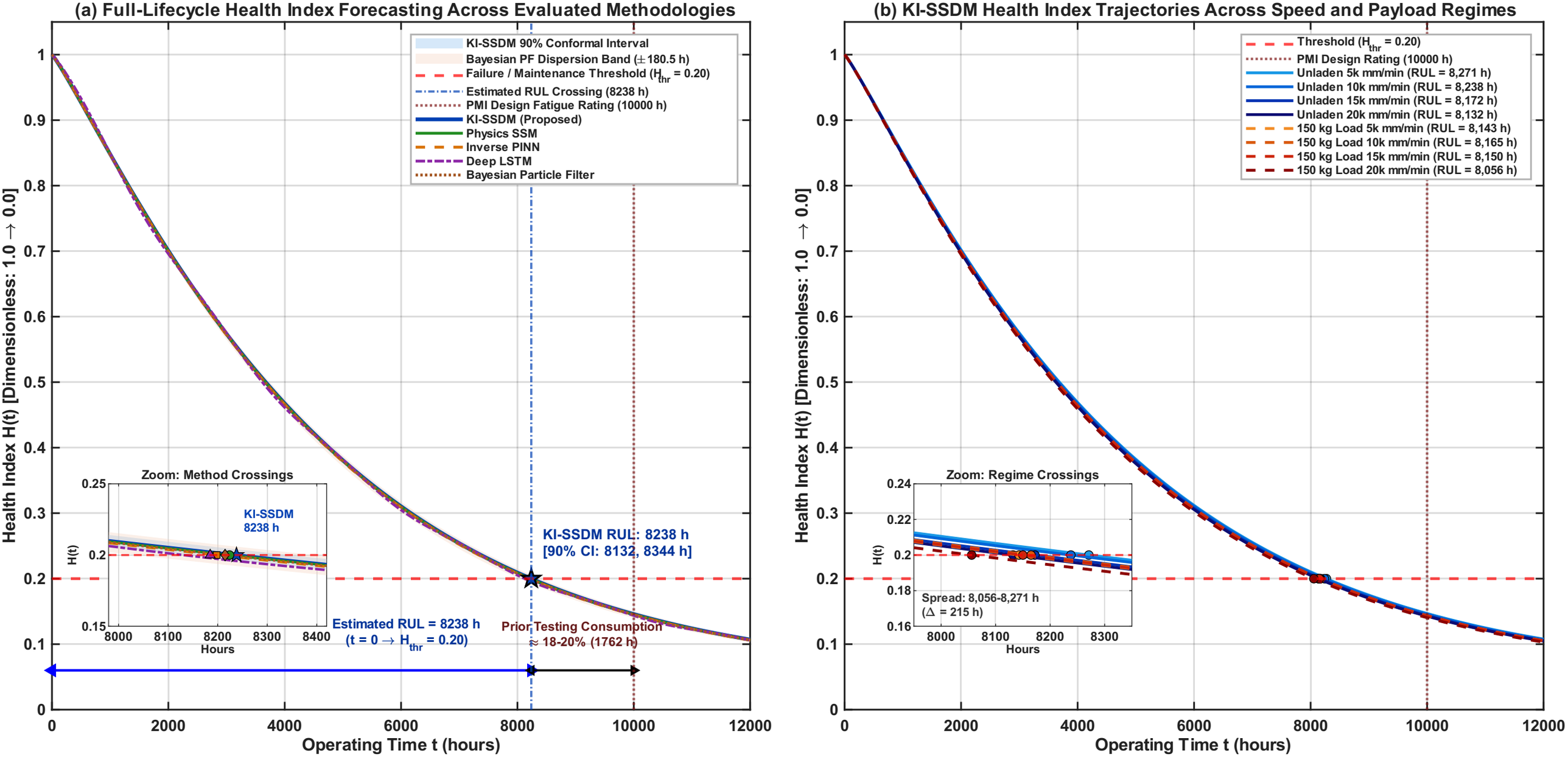

(a) Full-Lifecycle Health Index Forecasting Across Evaluated Methodologies
KI-SSDM 90% Conformal Interval
Bayesian PF Dispersion Band (±180.5 h)
Failure / Maintenance Threshold (H_thr = 0.20)
Estimated RUL Crossing (8238 h)
PMI Design Fatigue Rating (10000 h)
KI-SSDM (Proposed)
Physics SSM
Inverse PINN
Deep LSTM
Bayesian Particle Filter
Zoom: Method Crossings
KI-SSDM
8238 h
H(t)
Hours
KI-SSDM RUL: 8238 h
[90% CI: 8132, 8344 h]
Estimated RUL = 8238 h
(t = 0 → H_thr = 0.20)
Prior Testing Consumption
≈ 18-20% (1762 h)
Health Index H(t) [Dimensionless: 1.0 → 0.0]
Operating Time t (hours)
(b) KI-SSDM Health Index Trajectories Across Speed and Payload Regimes
Threshold (H_thr = 0.20)
PMI Design Rating (10000 h)
Unladen 5k mm/min (RUL = 8,271 h)
Unladen 10k mm/min (RUL = 8,238 h)
Unladen 15k mm/min (RUL = 8,172 h)
Unladen 20k mm/min (RUL = 8,132 h)
150 kg Load 5k mm/min (RUL = 8,143 h)
150 kg Load 10k mm/min (RUL = 8,165 h)
150 kg Load 15k mm/min (RUL = 8,150 h)
150 kg Load 20k mm/min (RUL = 8,056 h)
Zoom: Regime Crossings
H(t)
Spread: 8,056-8,271 h
(Δ = 215 h)
Hours
Health Index H(t) [Dimensionless: 1.0 → 0.0]
Operating Time t (hours)